

Week Summary

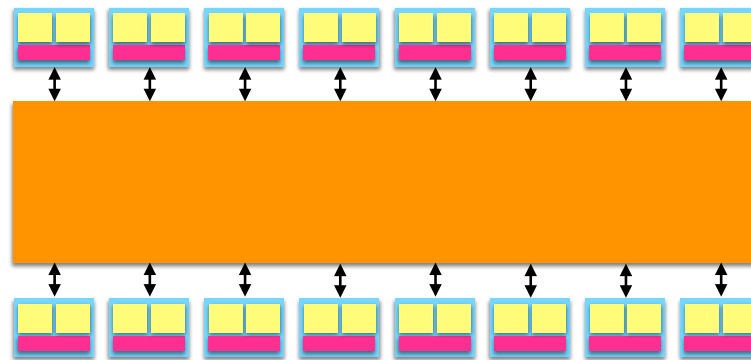
Introduction to OpenMP

Week Summary: Introduction to OpenMP

- The first four units showed how to use OpenMP directives, environment variables, and the runtime library
- The next three units were about doing useful work in parallel by parallelising loops
- We need to make sure our parallel program is correct by considering:
 - Variable scoping
 - Data races

OpenMP needs shared memory

- The OpenMP execution model only works when processors share memory
- In a cluster processor cores in the same compute node can address the same memory
- In a cluster processor cores in different compute nodes address different memory
- If we are using a cluster this means that we can only use OpenMP within a single compute node



OpenMP advantages

- With OpenMP we add directives our program to instruct the compiler to parallelise certain region
- Can incrementally parallelise existing code:
 - When working with an existing program identify time consuming regions (e.g. loops) and parallelise these first
- Can compile without OpenMP to recover serial functionality (conditional compilation)